

Online DAG Learning from Streaming Observational Data

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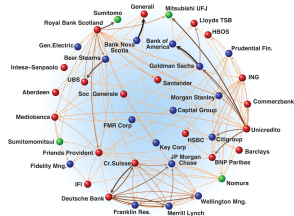
Collaborators: Gonzalo Mateos and Mariano Tepper

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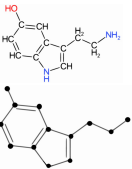
2026

- ▶ Graphs are natural models for relational data that help us learn in timely applications

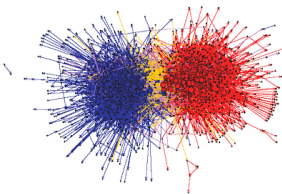
Economic Networks



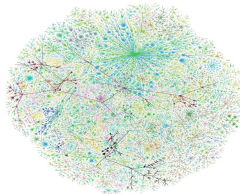
Molecules



Social and Information Networks



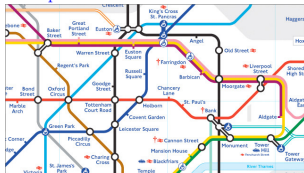
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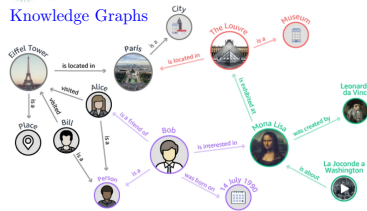
3D Meshes



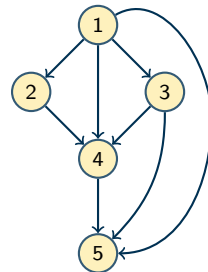
Transportation Networks



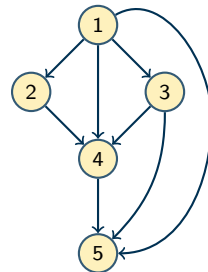
Knowledge Graphs



- ▶ **Goal:** learn the structure of **directed acyclic graphs (DAGs)** from observational data
- ▶ DAGs are prominent models across ML and the sciences
 - ⇒ Conditional independences among variables in Bayesian networks
 - ⇒ DAG edges may carry **causal interpretations**
 - ⇒ Bio [Sachs et al'05], finance [Sanford-Moosa'12], healthcare [Lucas'04]
- ▶ **Challenges:** directionality, acyclicity (combinatorial constraint), identifiability, NP-hard search [Chickering'96]



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- ▶ **This work:** most DAG learners are **batch** – they need the whole dataset up front. We develop an **online** framework that updates the DAG **as data streams in**, for both **static (SEM)** and **dynamic (SVAR)** models

Background: Linear causal models and DAG learning

Online problem formulation

Streaming SEM and SVAR algorithms

Adaptation to time-varying structure

Experiments

Conclusions

- ▶ DAG $\mathcal{G}(\mathcal{V}, \mathcal{E})$, vertices $\mathcal{V} = \{1, \dots, d\}$, weighted adjacency $\mathbf{W} \in \mathbb{R}^{d \times d}$
 $\Rightarrow W_{ij} \neq 0$ encodes a directed causal link $i \rightarrow j$
- ▶ Linear structural equation model (SEM): each node a noisy linear combination of its parents

$$\mathbf{X} = \mathbf{X}\mathbf{W} + \mathbf{Z}$$

- $\Rightarrow$ Rows of $\mathbf{X} \in \mathbb{R}^{n \times d}$ are i.i.d. samples; $\mathbf{Z}$ collects exogenous noise
- $\Rightarrow$ Equal variance (EV): $Z_{ij} \sim \mathcal{N}(0, \sigma^2)$ Non-equal (NV): $Z_{ij} \sim \mathcal{N}(0, \sigma_j^2)$

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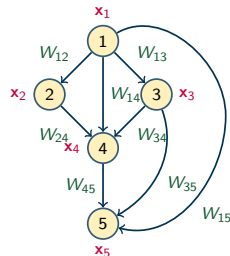
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- ▶ Structural vector autoregressive (SVAR): adds p lagged terms

$$\mathbf{x}_t = \mathbf{W}^\top \mathbf{x}_t + \sum_{k=1}^p \mathbf{A}_k^\top \mathbf{x}_{t-k} + \mathbf{z}_t \quad \Rightarrow \quad \mathbf{X} = \mathbf{X}\mathbf{W} + \mathbf{Y}\mathbf{A} + \mathbf{Z}$$

$\Rightarrow \mathbf{Y} = [\mathbf{X}_{-1}, \dots, \mathbf{X}_{-p}]$ stacks lags; $\mathbf{A} \in \mathbb{R}^{pd \times d}$ stacks $\{\mathbf{A}_k\}$

$\Rightarrow$ Acyclicity constrains $\mathbf{W}$ only – temporal order rules out cycles in $\mathbf{A}$



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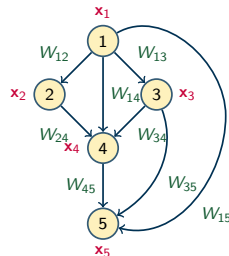
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$\Rightarrow$ Acyclicity constrains $\mathbf{W}$ only – temporal order rules out cycles in $\mathbf{A}$

- ▶ **Q**: estimate $\mathbf{W}$ (and $\mathbf{A}$) from data revealed sequentially over time?



- ▶ Recovering $\mathbf{W}$ from data is **NP-hard** as a search over DAGs [Chickering'96]
 - ⇒ **Key idea** [NOTEARS, Zheng et al'18]: replace the combinatorial constraint with a **smooth acyclicity function** $h(\mathbf{W}) = 0 \iff \mathbf{W}$ acyclic

$$\min_{\mathbf{W} \in \mathbb{R}^{d \times d}} \ell(\mathbf{W}; \mathbf{X}) + \lambda \|\mathbf{W}\|_1 \quad \text{s.t.} \quad h(\mathbf{W}) = 0$$

- ▶ **Acyclicity functions:**
 - ⇒ NOTEARS: $h_{\text{NT}}(\mathbf{W}) = \text{Trace}(e^{\mathbf{W} \odot \mathbf{W}}) - d$
 - ⇒ **DAGMA** [Bello et al'22]: $h_s(\mathbf{W}) = -\log \det(s\mathbf{I}_d - \mathbf{W} \odot \mathbf{W}) + d \log s$ – better gradients

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- ▶ **CoLiDE** [Saboksayr et al'24]: jointly estimate the **noise scale** σ with $\mathbf{W}$

$$S(\mathbf{W}, \sigma; \mathbf{X}) = \frac{1}{2n\sigma} \|\mathbf{X} - \mathbf{X}\mathbf{W}\|_F^2 + \frac{d\sigma}{2} + \lambda \|\mathbf{W}\|_1$$

⇒ Decouples λ from σ ; closed-form σ update; handles heteroscedastic noise (NV)

- ▶ NOTEARS, GOLEM, DAGMA, CoLiDE, Dynotears, PCMCi ... all assume the **full dataset is available a priori** and revisited every iteration
- ▶ But many modern processes generate data **sequentially**: financial markets, sensor networks, industrial monitoring, biological dynamics [time series]
- ▶ **Retraining from scratch** on each new sample is computationally wasteful – and the underlying graph may **drift over time**

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- ▶ **This work – an online DAG learner that:**

- ⇒ processes observations in **mini-batches**, including the true-online $\text{batch_size} = 1$ regime
- ⇒ handles **both SEM and SVAR** generative models
- ⇒ keeps per-step cost **independent** of the cumulative sample count
- ⇒ **tracks** time-varying DAGs through abrupt structural change

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- ▶ Sequential protocol indexed by $t = 1, 2, \dots$; at time t the learner receives a batch

$$\mathbf{X}_t \in \mathbb{R}^{n_t \times d}, \quad N_t = \sum_{\tau=1}^t n_\tau \text{ (cumulative size)}$$

$\Rightarrow n_t = 1$: true-online regime (one sample per step)

$\Rightarrow n_t > 1$: mini-batch streaming (e.g. periodic sensor / financial readouts)

- ▶ After processing $\mathbf{X}_t$, produce an updated estimate $\widehat{\mathbf{W}}_t$ **without revisiting** the history $\mathbf{X}^{(t-1)}$

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- ▶ **Operational desiderata**

$\Rightarrow$ (1) **Bounded per-step cost**: update of $\widehat{\mathbf{W}}_t$ **must not grow with t**

$\Rightarrow$ (2) **Consistency with the batch limit**: as $t \rightarrow \infty$ (stationary), match the batch fixed point

$\Rightarrow$ (3) **Adaptivity**: recover a new structure within bounded delay after a change

- **Streaming SEM.** At each t , minimize the population risk using only $\{\mathbf{X}_\tau\}_{\tau \leq t}$:

$$\mathcal{L}(\mathbf{W}, \sigma) = \mathbb{E} \left[\frac{1}{2\sigma} \|\mathbf{x} - \mathbf{W}^\top \mathbf{x}\|_2^2 \right] + \frac{d\sigma}{2} + \lambda \|\mathbf{W}\|_1 \quad \text{s.t.} \quad h_s(\mathbf{W}) = 0$$

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- **Streaming SVAR.** Jointly estimate $(\mathbf{W}, \mathbf{A})$:

$$\mathcal{L}(\mathbf{W}, \mathbf{A}, \sigma) = \mathbb{E} \left[\frac{1}{2\sigma} \|\mathbf{x}_t - \mathbf{W}^\top \mathbf{x}_t - \mathbf{A}^\top \mathbf{y}_t\|_2^2 \right] + \frac{d\sigma}{2} + \lambda_{\mathbf{W}} \|\mathbf{W}\|_1 + \lambda_{\mathbf{A}} \|\mathbf{A}\|_1$$

⇒ $\mathbf{y}_t = [\mathbf{x}_{t-1}^\top, \dots, \mathbf{x}_{t-p}^\top]^\top$; separate penalties since temporal effects in **A** are weaker

- ▶ In nonstationary environments the DAG **itself evolves** over time

$$\mathbf{W}_t^* = \mathbf{W}_{t-1}^* + \mathbf{\Delta}_t$$

where $\mathbf{\Delta}_t$ is a **sparse perturbation** that may add, remove, or flip edges while preserving acyclicity

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- ▶ **Goal shifts from convergence to tracking:** the estimate $\widehat{\mathbf{W}}_t$ should remain close to the **active** ground truth $\mathbf{W}_t^*$ along the trajectory

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- Write the CoLiDE score in terms of the data **covariance** $\Sigma = \mathbb{E}[\mathbf{x}\mathbf{x}^\top]$:

$$S(\mathbf{W}, \sigma; \Sigma) = \frac{1}{2\sigma} \text{Trace}((\mathbf{I}_d - \mathbf{W})^\top \Sigma (\mathbf{I}_d - \mathbf{W})) + \frac{d\sigma}{2}$$

⇒ The score depends on the data **only through** Σ – the engine of the streaming algorithm

- Relax acyclicity via a sequence of weighted penalty subproblems $i = 0, \dots, T-1$:

$$(\widehat{\mathbf{W}}^{(i)}, \widehat{\sigma}^{(i)}) \in \arg \min_{\mathbf{W}, \sigma} \mu_i [S(\mathbf{W}, \sigma; \Sigma) + \lambda \|\mathbf{W}\|_1] + h_{s_i}(\mathbf{W})$$

⇒ $\mu_i = \mu_0 \rho^i$ ($\rho \in (0, 1)$) decays geometrically; $\{s_i\}$ annealed; **warm-start** from previous iterate

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- ▶ **Key departure:** Σ is unknown – replace it by a **running, incrementally-updated proxy** Σ_t

► **Running covariance** Σ_t , with $\Sigma_0 = \mathbf{0}$

⇒ **Mini-batch** ($n_t > 1$), convex combination:

$$\Sigma_t = \frac{t-1}{t} \Sigma_{t-1} + \frac{1}{t} \frac{1}{n_t} \tilde{\mathbf{X}}_t^\top \tilde{\mathbf{X}}_t$$

⇒ **True-online** ($n_t = 1$), **Welford** update:

$$\mu_t = \mu_{t-1} + \frac{1}{t} (\mathbf{x}_t - \mu_{t-1})$$

$$\mathbf{M}_t = \mathbf{M}_{t-1} + (\mathbf{x}_t - \mu_{t-1})(\mathbf{x}_t - \mu_{t-1})^\top$$

$$\Sigma_t = \mathbf{M}_t / t$$

⇒ Both cost $\mathcal{O}(d^2)$ per step, independent of N_t

► **Gradient on $\mathbf{W}$**

$$\nabla_{\mathbf{W}} S = -\frac{1}{\sigma} \Sigma_t (\mathbf{I}_d - \mathbf{W})$$

$$\mathbf{G}_{\mathbf{W}} = \mu_i \left(-\frac{1}{\sigma} \Sigma_t (\mathbf{I}_d - \mathbf{W}) + \lambda \text{sign}(\mathbf{W}) \right) + 2\mathbf{W} \odot \mathbf{M}^{-\top}$$

with $\mathbf{M} = s_i \mathbf{I}_d - \mathbf{W} \odot \mathbf{W}$; one **Adam** step on $\mathbf{W}$

⇒ **Feasibility safeguard**: roll back & halve the step if $\mathbf{M}$ loses positive-definiteness

► **Closed-form noise scale**

⇒ **EV**:

$$\hat{\sigma}_t = \sqrt{\frac{1}{d} \text{Trace}((\mathbf{I}_d - \mathbf{W})^\top \Sigma_t (\mathbf{I}_d - \mathbf{W}))}$$

⇒ **NV**: $\hat{\sigma}_{t,j}^2 = [(\mathbf{I}_d - \mathbf{W})^\top \Sigma_t (\mathbf{I}_d - \mathbf{W})]_{jj}$

⇒ **NV reuses the same Σ_t** – diagonal instead of trace

Input: stream $\{\mathbf{X}_t\}$, sparsity λ , schedules $\{\mu_i\}, \{s_i\}$, inner iters K_i , rate η

Output: $\hat{\mathbf{W}}$

$\mathbf{W}_0 \leftarrow \mathbf{0}, \sigma_0 \leftarrow 1, \Sigma_0 \leftarrow \mathbf{0}$

for $i = 0, \dots, T-1$ **do**

 reset Adam state for $\mathbf{W}$

for $k = 1, \dots, K_i$ **do**

 observe $\mathbf{X}_t$, center to $\tilde{\mathbf{X}}_t$

 update Σ_t (convex comb. or Welford)

 compute $\mathbf{G}_\mathbf{W}$

$\mathbf{W} \leftarrow \mathbf{W} - \eta \cdot \text{Adam}(\mathbf{G}_\mathbf{W})$

 update σ (EV or NV closed form)

return $\mathbf{W}$

► **Per-step cost**

$\Rightarrow \mathcal{O}(d^3)$ for the inverse
 $\mathbf{M}^{-1}$

$\Rightarrow \mathcal{O}(n_t d^2)$ covariance
 update

$\Rightarrow \mathcal{O}(d^2)$ memory,
 independent of N_t

► **Consistency:** $\Sigma_t \rightarrow \Sigma$ + batch
fixed-point convergence $\Rightarrow$
streaming matches batch as
 $t \rightarrow \infty$

- Now the score depends on **three** second-order statistics, all tracked online:

$$\boldsymbol{\Sigma}_{XX} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}, \quad \boldsymbol{\Sigma}_{XY} = \frac{1}{n} \mathbf{X}^\top \mathbf{Y}, \quad \boldsymbol{\Sigma}_{YY} = \frac{1}{n} \mathbf{Y}^\top \mathbf{Y}$$

$$S = \frac{1}{2\sigma} \left[\text{Trace}((\mathbf{I}_d - \mathbf{W})^\top \boldsymbol{\Sigma}_{XX} (\mathbf{I}_d - \mathbf{W})) - 2 \text{Trace}((\mathbf{I}_d - \mathbf{W})^\top \boldsymbol{\Sigma}_{XY} \mathbf{A}) + \text{Trace}(\mathbf{A}^\top \boldsymbol{\Sigma}_{YY} \mathbf{A}) \right] + \frac{d\sigma}{2}$$

⇒ Updated block-wise by the same convex combination (Welford on $[\mathbf{x}_t^\top, \mathbf{y}_t^\top]^\top$ for $n_t = 1$)

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- **Lag buffering across batch seams.** A circular buffer $\mathcal{B}_t \in \mathbb{R}^{p \times d}$ of the last p rows:

$$\mathcal{B}_t \leftarrow \text{last } p(\mathcal{B}_{t-1} \parallel \mathbf{X}_t)$$

⇒ First p rows of $\mathbf{X}_t$ pair with lags from $\mathcal{B}_{t-1}$ – **no row pair dropped**, at $\mathcal{O}(pd)$ extra memory

► Gradients from running statistics

$$\nabla_{\mathbf{W}} S = \frac{1}{\sigma} (\boldsymbol{\Sigma}_{XY,t} \mathbf{A} - \boldsymbol{\Sigma}_{XX,t} (\mathbf{I}_d - \mathbf{W}))$$

$$\nabla_{\mathbf{A}} S = \frac{1}{\sigma} (\boldsymbol{\Sigma}_{YY,t} \mathbf{A} - \boldsymbol{\Sigma}_{XY,t}^\top (\mathbf{I}_d - \mathbf{W}))$$

⇒ $\mathbf{G}_{\mathbf{W}}$ adds $\lambda_{\mathbf{W}} \text{sign}(\mathbf{W})$ and $2\mathbf{W} \odot \mathbf{M}^{-\top}$

⇒ $\mathbf{G}_{\mathbf{A}}$ adds $\lambda_{\mathbf{A}} \text{sign}(\mathbf{A})$ – **no acyclicity term**

⇒ Two independent Adam states for $\mathbf{W}$ and $\mathbf{A}$

► **EV** σ reads the trace of the 3-term residual covariance; **NV** reads its diagonal

► **Cost**: $\mathcal{O}(d^3)$ inverse + $\mathcal{O}((d+pd)^2 n_t)$ covariance; memory $\mathcal{O}((d+pd)^2) + \mathcal{O}(pd)$, independent of t

Input: stream, lag p , $\lambda_{\mathbf{W}}$, $\lambda_{\mathbf{A}}$, schedules, K_i , η

Output: $\hat{\mathbf{W}}, \hat{\mathbf{A}}$

init $\mathbf{W}, \mathbf{A}, \sigma$; covariances $\leftarrow \mathbf{0}$; buffer $\mathcal{B} \leftarrow \emptyset$

for $i = 0, \dots, T-1$ **do**

 reset Adam states for $\mathbf{W}, \mathbf{A}$

for $k = 1, \dots, K_i$ **do**

 observe $\mathbf{X}_t$; form $(\mathbf{X}_t, \mathbf{Y}_t)$ via $\mathcal{B}$

 update $\mathcal{B}$; update $\boldsymbol{\Sigma}_{XX}, \boldsymbol{\Sigma}_{XY}, \boldsymbol{\Sigma}_{YY}$

 compute $\mathbf{G}_{\mathbf{W}}, \mathbf{G}_{\mathbf{A}}$

 Adam step on $\mathbf{W}$ and on $\mathbf{A}$

 update σ (EV / NV)

return $\mathbf{W}, \mathbf{A}$

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- ▶ All-history covariance gives the newest batch **vanishing weight** – great under stationarity, destructive under change
- ▶ **Fix:** recompute the covariance on a sliding window of the last W observations

$$\Sigma_t^{(W)} = \frac{1}{|\mathcal{W}_t|} \sum_{\mathbf{x} \in \mathcal{W}_t} (\mathbf{x} - \bar{\mathbf{x}}_{\mathcal{W}_t})(\mathbf{x} - \bar{\mathbf{x}}_{\mathcal{W}_t})^\top, \quad \mathcal{W}_t = \{\mathbf{X}_{t-W+1}, \dots, \mathbf{X}_t\}$$

- ⇒ Refit the augmented Lagrangian on the window, **warm-started** from $\widehat{\mathbf{W}}_{t-1}$
- ⇒ W is the **tracking-variance knob**: small W adapts fast but noisier; large W smooths but lags $\Theta(W)$

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- ⇒ W is the **tracking-variance knob**: small W adapts fast but noisier; large W smooths but lags $\Theta(W)$
- ▶ **Three-phase behavior** around a change point t^* :
 - ⇒ **Pre-change** ($t < t^*$): window all old $\Rightarrow$ converges to $\mathbf{W}_1^*$
 - ⇒ **Transition** ($t^* \leq t < t^* + W$): mixed window $\Rightarrow$ intermediate estimates
 - ⇒ **Post-adaptation** ($t \geq t^* + W$): window all new $\Rightarrow$ converges to $\mathbf{W}_2^*$
 - ⇒ **Adaptation delay** $\approx W$ batches – no external change-detector needed

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Conclusions

- ▶ **Graphs:** Erdős–Rényi DAGs, $d \in \{20, 40, 60, 80, 100\}$ (SEM) / $\{10, \dots, 50\}$ (SVAR), densities ER2/ER3/ER4
 - ⇒ Edge weights $\sim U(-2, -0.5) \cup (0.5, 2)$; SVAR lag $p = 1$, temporal strength 0.5 (companion-stable)
- ▶ **Data:** $n = 1000$ samples / time points; EV unit-variance Gaussian or NV variances $\sim U(0.5, 10)$
- ▶ **Methods:** full-batch STATIC reference vs. streaming at $\text{batch_size} \in \{100, 50, 1\}$
 - ⇒ Identical schedule: $T = 4$ outer, $K_{\text{warm}} = 20\text{k}$, $K_{\text{final}} = 70\text{k}$, $\eta = 3 \times 10^{-4}$
- ▶ **Metrics:** SHD (edge flips) and $\text{NMSE} = \|\widehat{\mathbf{W}} - \mathbf{W}^*\|_F^2 / \|\mathbf{W}^*\|_F^2$; mean $\pm$ std over 5 seeds
- ▶ **Four questions:** (i) does mini-batch streaming match full-batch on SEM? (ii) same for SVAR? (iii) sensitivity to $(\lambda_{\mathbf{W}}, \lambda_{\mathbf{A}})$? (iv) can the sliding window track a drifting DAG?

Table: SHD under EV noise (mean $\pm$ std, 5 seeds). Streaming variants are statistically indistinguishable from STATIC.

Method	$d=20$	$d=40$	$d=60$	$d=80$	$d=100$
<i>ER2</i>					
STATIC	0.0 ± 0.0	1.6 ± 1.6	2.0 ± 2.6	3.0 ± 2.5	3.6 ± 2.4
bs=1	0.0 ± 0.0	0.8 ± 1.2	2.8 ± 4.2	2.8 ± 2.5	3.6 ± 2.4
<i>ER4</i>					
STATIC	6.0 ± 6.0	7.8 ± 6.0	23.2 ± 12.8	15.0 ± 9.9	38.2 ± 28.3
bs=100	5.0 ± 5.7	17.4 ± 22.3	28.6 ± 24.5	14.0 ± 10.3	41.8 ± 27.8
bs=1	8.0 ± 9.6	11.0 ± 5.8	19.0 ± 12.2	14.8 ± 9.7	38.0 ± 27.3

- All four methods overlap within one standard deviation at every cell – including the true-online bs = 1 regime
 - ⇒ NV recovery (e.g. ER4, $d=100$: 108.6 vs. 111.4 vs. 113.4 vs. 113.4) is harder but streaming still tracks Static
 - ⇒ Sparsity tuning: SHD-optimal λ is the same across all four methods – shared loss landscape

Table: SVAR recovery at $d=20$, ER2, lag $p=1$, $T_{\text{steps}}=1000$ (mini-batch SGD). Mean $\pm$ std.

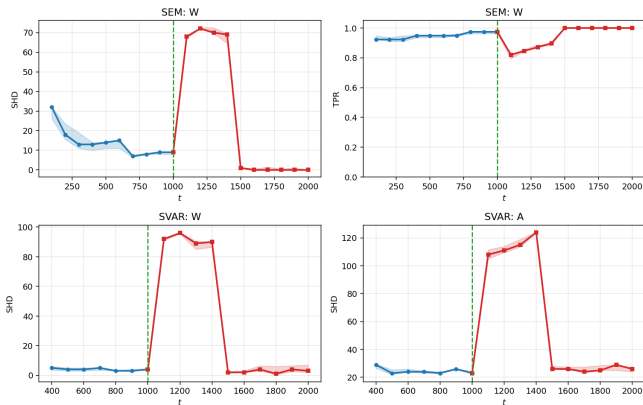
Method	W SHD	A SHD	W NMSE	A NMSE
<i>EV ($\lambda_W=0.01$, $\lambda_A=0.015$; 5 seeds)</i>				
STATIC (EV)	2.4 ± 1.4	14.2 ± 5.8	0.006	0.262
bs=100	1.8 ± 1.2	13.2 ± 6.1	0.004	0.247
bs=1	2.0 ± 1.6	13.2 ± 6.1	0.004	0.248
<i>NV ($\lambda_W=0.08$, $\lambda_A=0.05$; 10 seeds)</i>				
bs=100	13.3 ± 6.9	19.9 ± 6.3	0.163	0.412
bs=1	14.0 ± 8.1	20.2 ± 6.1	0.174	0.422

► Contemporaneous **W** recovered near-exactly; the three mini-batch variants agree within **one SHD** edge on both sides

⇒ **A** is harder: pd^2 free parameters, **no acyclicity** constraint, and SVAR stability pushes weak true edges below threshold

⇒ Under NV, full-batch STATIC **fails to converge** at any λ – yet the streaming variants still find a stable fixed point

Results: tracking a mid-stream DAG change



► $d=20$, ER2, batch=100, window $W=500$, change point $t^*=1000$; median over 3 realizations

⇒ **SEM:** W-SHD converges to 9 pre-change, spikes to 72 during window mixing, drops to 0 (TPR = 1.00) at $t = 1500$

⇒ **SVAR:** W-SHD settles at 5 → 3; A-SHD floors 23 → 26 (threshold artifact, not adaptation failure)

⇒ Transition spans exactly $\Theta(W/\text{batch}) = 5$ batches – matching the analysis

Background: Linear causal models and DAG learning

Online problem formulation

Streaming SEM and SVAR algorithms

Adaptation to time-varying structure

Experiments

Conclusions

- ▶ Proposed an **online DAG learning framework** for streaming data, for both **static (SEM)** and **dynamic (SVAR)** models
 - ⇒ Augments the **CoLiDE** family with **running second-order statistics** – one covariance for SEM, the joint block $(\Sigma_{XX}, \Sigma_{XY}, \Sigma_{YY})$ for SVAR
 - ⇒ Mini-batch SGD inner loop: $\mathcal{O}(d^2)$ **memory**, $\mathcal{O}(d^3)$ **time** per step, independent of cumulative samples
 - ⇒ Closed-form noise updates (EV & NV); Welford backs the true-online $bs = 1$ regime
 - ⇒ **Sliding-window** variant tracks evolving DAGs with delay $\Theta(W/\text{batch})$
 - ⇒ **Streaming = batch** in recovery quality at convergence – differing only in per-step cost and adaptivity

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- ▶ **Ongoing and future work:**
 - ⇒ **Regret analysis:** $\mathcal{O}(\sqrt{T})$ static and dynamic regret under a path-length budget on $\{\mathbf{W}_t^*\}$
 - ⇒ **Incremental warm-starts:** reuse Adam moments & schedule index to cut per-batch cost
 - ⇒ **Data-driven thresholds** and online $(\lambda_{\mathbf{W}}, \lambda_{\mathbf{A}})$ schedules
 - ⇒ **Beyond linear/Gaussian:** LiNGAM-style and nonlinear (neural) SEMs; **real-data** streaming evaluation